

Equality in the Fradelizi–Paouris–Schütt second-moment inequality

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Abstract

Fradelizi, Paouris, and Schütt proved a sharp lower bound for the second moment of a convex body whose extreme points lie outside an open Euclidean ball. They characterized equality for polytopes and conjectured the same characterization for arbitrary convex bodies. We prove this conjecture: equality holds precisely for nondegenerate simplices whose vertices lie on the bounding sphere. The proof extends their simplex argument by an almost-everywhere partition into countably many simplices with extreme-point vertices. An exact nonnegative deficit identity retains the equality information for this partition.

1 Statement and relation to the original problem

A convex body is a compact convex subset of $\mathbb{R}^n$ with nonempty interior. Write $|K|$ for its n -dimensional volume and define

$$g_K = \frac{1}{|K|} \int_K x \, dx, \quad C_2(K) = \frac{1}{|K|} \int_K \|x\|_2^2 \, dx.$$

The inequality below and its equality characterization for polytopes are [1, Theorem 1.1]. The conjecture immediately following that theorem asks for the same equality characterization for all convex bodies. It was also recorded in [2, p. 2908], and is the question catalogued as OWR-4136-011 in [3]. Our contribution is the general-body equality case; the inequality and the simplex moment formula are due to [1].

Theorem 1. *Let $n \geq 1$, $r > 0$, and let $K \subset \mathbb{R}^n$ be a convex body satisfying $\|v\|_2 \geq r$ for every $v \in \text{ext } K$. Then*

$$C_2(K) \geq \frac{r^2 + (n+1)\|g_K\|_2^2}{n+2}. \quad (1)$$

Equality holds if and only if K is a nondegenerate n -simplex and all its vertices have norm r .

The norm hypothesis is explicitly non-strict: the original sources’ prose uses “greater than,” while their equality statement includes vertices of norm r . Under the strictly greater hypothesis equality is impossible. Neither $g_K = 0$ nor $0 \in K$ is assumed. The catalogue asks the question for $n > 2$; the theorem includes these dimensions and also $n = 1, 2$.

2 An extreme-point simplex partition

Lemma 2. *Every convex body $K \subset \mathbb{R}^n$ admits a finite or countable family (S_j) of nondegenerate n -simplices with pairwise disjoint interiors, vertices in $\text{ext } K$, and*

$$\left| K \setminus \bigcup_j S_j \right| = 0.$$

The construction may start with any nondegenerate simplex whose vertices are extreme points of K .

Proof. The finite-dimensional extreme-point theorem gives $K = \overline{\text{conv}(\text{ext } K)}$. Thus $\text{ext } K$ affinely spans $\mathbb{R}^n$. Choose distinct extreme points $v_1, \dots, v_{n+1}$ that are affinely independent. Because every subspace of Euclidean space is second countable, $\text{ext } K$ has a finite or countable dense subset D containing these points. Enumerate D without repetition, with these points first, and set

$$P_m = \text{conv}(v_1, \dots, v_m), \quad m \geq n+1.$$

If D is finite, stop at its last point. A subsequently inserted point $v = v_{m+1}$ is outside P_m , since otherwise it would be a convex combination of other points of K , contradicting extremality.

We recall explicitly the finite shell construction. Use a facet representation

$$P_m = \{x : a_F \cdot x \leq b_F \text{ for every facet } F\}.$$

Call F visible from v if $a_F \cdot v > b_F$. Then

$$P_{m+1} = P_m \cup \bigcup_{F \text{ visible}} \text{conv}(v, F). \quad (2)$$

Indeed, for $x \in P_{m+1} \setminus (P_m \cup \{v\})$, the ray from v through x first meets P_m at a point y lying on a visible facet. Otherwise every facet inequality active at y would hold at v ; all inactive inequalities would still be strict just before y , contradicting first intersection. Also $x \in [v, y]$. The apex v is covered because at least one visible facet exists. Distinct visible pyramids have disjoint interiors: an interior point determines a unique first intersection of its ray with P_m , in the relative interior of its base facet. Their interiors lie outside P_m .

A finite polytope can be partitioned into simplices using only its vertices. To see this, induct on dimension, choosing a vertex and coning it to vertex partitions of all facets not containing that vertex. Rays through interior points give disjoint interiors and coverage; closure gives the boundary. The base case is a point. Apply this construction to every visible facet in (2), and cone its $(n-1)$ -simplices to v . Each resulting n -simplex is nondegenerate and has vertices in D . Starting with P_{n+1} and retaining every earlier simplex gives a finite or countable family with disjoint interiors and union $C = \bigcup_m P_m = \text{conv } D$. Compatibility of subdivisions along lower-dimensional faces is unnecessary.

Density gives $\overline{C} = K$. The convex set C has nonempty interior, so $\text{int } C = \text{int } \overline{C} = \text{int } K$. One can see the nontrivial inclusion by surrounding any interior point of $\overline{C}$ with a small simplex and approximating its vertices by points of C ; a sufficiently small perturbation still surrounds that point. Thus $K \setminus C$ is contained in ∂K , which has zero n -dimensional volume. $\square$

3 The deficit identity and equality

For a nondegenerate simplex $S = \text{conv}(x_0, \dots, x_n)$ put $c_S = (n+1)^{-1} \sum_i x_i$ and $q_S = (n+1)^{-1} \sum_i \|x_i\|_2^2$. Uniform barycentric coordinates satisfy

$$\mathbb{E} \lambda_i^2 = \frac{2}{(n+1)(n+2)}, \quad \mathbb{E} \lambda_i \lambda_k = \frac{1}{(n+1)(n+2)} \quad (i \neq k).$$

These follow by integrating monomials on the standard simplex; more generally $\mathbb{E} \prod_i \lambda_i^{a_i} = n! \prod_i a_i! / (n + \sum_i a_i)!$ for nonnegative integers a_i . Expanding $\|\sum_i \lambda_i x_i\|_2^2$ gives the identity of [1, Lemma 3.2]:

$$C_2(S) = \frac{\sum_i \|x_i\|_2^2 + \|\sum_i x_i\|_2^2}{(n+1)(n+2)} = \frac{q_S + (n+1)\|c_S\|_2^2}{n+2}. \quad (3)$$

Proof of Theorem 1. Use Lemma 2, and let $w_j = |S_j|/|K|$, $c_j = c_{S_j}$, and $q_j = q_{S_j}$. All weights are positive, and null overlaps and countable additivity give

$$\sum_j w_j = 1, \quad \sum_j w_j c_j = g_K, \quad C_2(K) = \sum_j w_j C_2(S_j).$$

All points and centroids lie in the bounded set K , so every series used here and below is absolutely convergent. Each $q_j \geq r^2$. Applying (3) and the weighted variance identity yields

$$\begin{aligned} \Delta_r(K) &:= C_2(K) - \frac{r^2 + (n+1)\|g_K\|_2^2}{n+2} \\ &= \frac{1}{n+2} \sum_j w_j (q_j - r^2) + \frac{n+1}{n+2} \sum_j w_j \|c_j - g_K\|_2^2. \end{aligned} \quad (4)$$

Both sums are nonnegative, proving (1). If equality holds, every positive-weight summand vanishes. Thus $q_j = r^2$ and $c_j = g_K$ for every j . The first condition forces every vertex of every S_j to have norm r . The second condition forces the family to consist of one simplex: the centroid of a nondegenerate simplex is in its interior, and two simplices with the same centroid cannot have disjoint interiors. The union in the construction is dense in K , so the sole closed simplex is K . Conversely, an inscribed simplex has $q_S = r^2$, and (3) gives equality. $\square$

4 A finite witness to strictness

The preceding proof gives a useful quantitative check on the passage from polytopes to arbitrary bodies. Start with an extreme-point simplex $S = \text{conv}(v_0, \dots, v_n)$. If $K \neq S$, there is an extreme point $u \notin S$. Insert u next. Choose a strictly visible facet opposite v_i and put $T = \text{conv}(u, \{v_j : j \neq i\})$. Both S and T are retained in the partition, have positive volumes a, b , and satisfy $c_T - c_S = (u - v_i)/(n + 1)$. Completing the square in their two variance terms in (4) gives

$$\Delta_r(K) \geq \frac{ab \|u - v_i\|_2^2}{(n + 1)(n + 2)|K|(a + b)} > 0. \quad (5)$$

Equivalently, apply the finite deficit identity on each P_m retaining S, T , replace $|P_m|$ by the larger $|K|$, and pass to the limit in bounded volume integrals. This supplies a second route to equality rigidity without assuming a countable partition in advance.

Corollary 3. *Under the hypotheses of Theorem 1, equality in $C_2(K) \geq r^2/(n + 2)$ holds exactly for simplices with all vertices of norm r and centroid zero.*

For the strengthened inequality (1), regularity is never required. For example, the non-regular triangle with vertices $(1, 0), (0, 1), (-1, 0)$ has $g_K = (0, 1/3)$ and $C_2(K) = 1/3$, giving equality with $r = 1$. In dimension one the equality interval is $[-r, r]$.

Verification and provenance. The proof is analytic and does not depend on computation. The accompanying standard-library Python script checks simplex moments by exact polynomial integration and finite shell identities by independent polygon moment formulas. The package also contains a separate finite-approximation proof and adversarial audit notes. These checks are not a proof-assistant formalization. A bounded literature audit completed on 23 September 2026 found no earlier general-body equality resolution in the sources examined; it does not certify priority. This unrefereed, AI-assisted manuscript was prepared with OpenAI Codex from a candidate supplied by the author from prior AI-assisted work, with separate AI verification agents.

References

- [1] M. Fradelizi, G. Paouris, and C. Schütt, *Simplices in the Euclidean ball*, Canad. Math. Bull. **55** (2012), no. 3, 498–508. [doi:10.4153/CMB-2011-142-1](https://doi.org/10.4153/CMB-2011-142-1).
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